

Introduction to Fourier Transform

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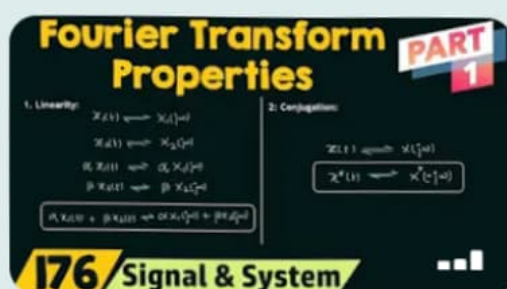
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Conditions for Existence of Fourier Transform (Dirichlet...

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Properties of Fourier Transf...

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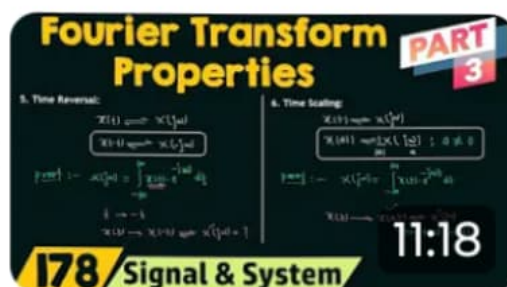
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Properties of Fourier Transform (Part 2)

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Properties of Fourier Transform (Part 3)

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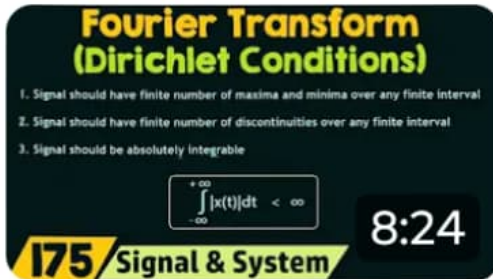
Properties of Fourier

Fourier Transform | Chapte... · 4/49

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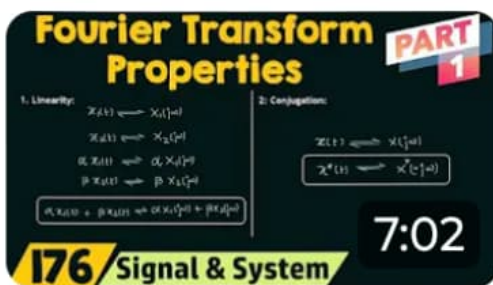
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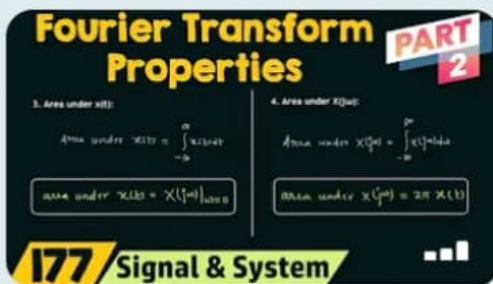
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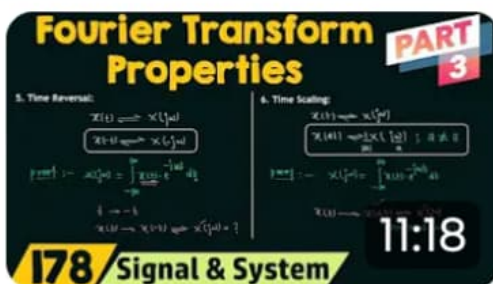
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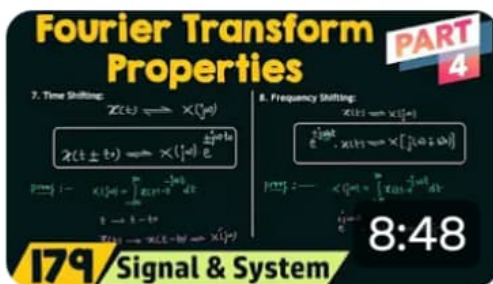
Properties of Fourier Transform (Part 3)

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Properties of Fourier Transform (Part 4)

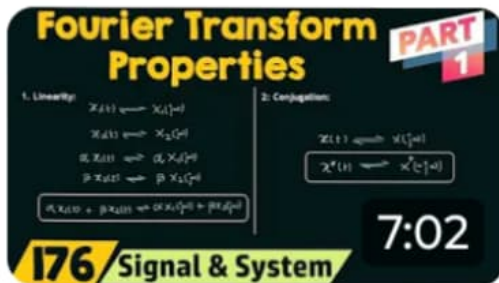
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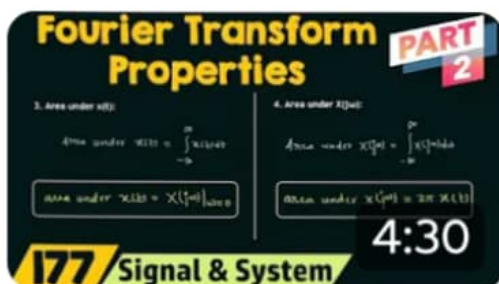




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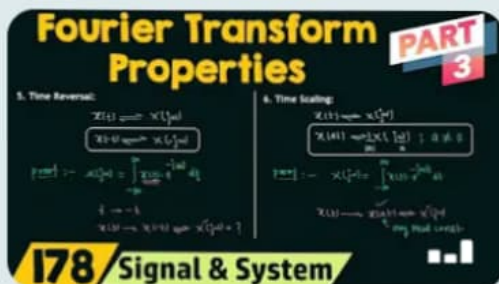
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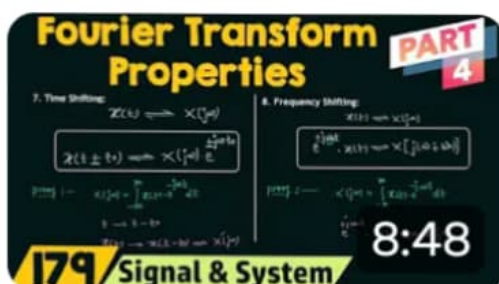
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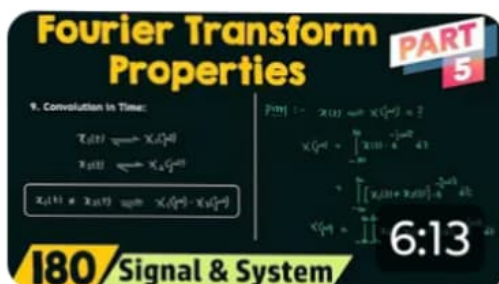
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Properties of Fourier Transform (Part 5)

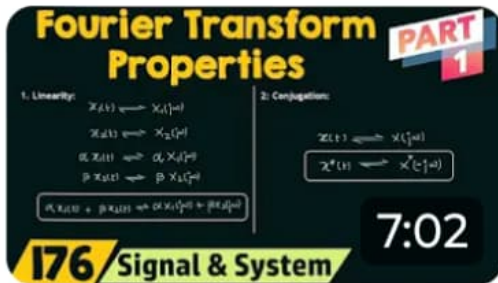
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Properties of Fourier Transform (Part 1)

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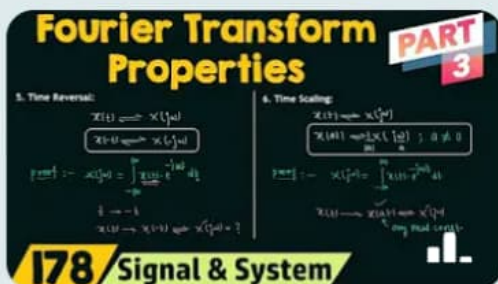
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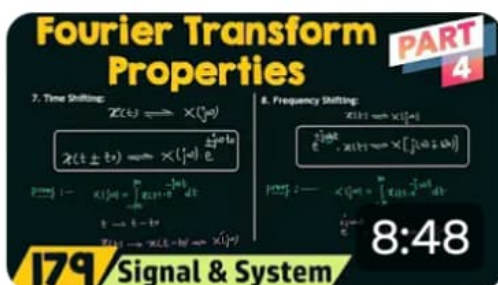
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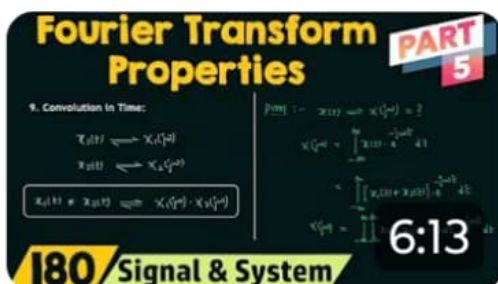
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Properties of Fourier Transform (Part 5)

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Properties of Fourier Transform (Part 3)

5. Time Reversal:

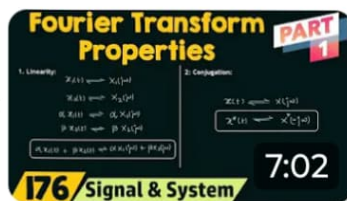
$$\begin{aligned}
 x(t) &\Leftrightarrow X(j\omega) \\
 x(-t) &\Leftrightarrow X(-j\omega) \\
 \text{proof :- } X(j\omega) &= \int_{-\infty}^{\infty} x(t) \cdot e^{-j\omega t} dt \\
 t &\rightarrow -t \\
 x(t) &\rightarrow x(-t) \Rightarrow X(j\omega) = ? \\
 X(j\omega) &= \int_{-\infty}^{\infty} x(-t) \cdot e^{-j\omega t} dt \\
 -t = \tau &\Rightarrow t = -\tau \Rightarrow dt = -d\tau \\
 t = -\infty &\Rightarrow \tau = \infty \\
 t = \infty &\Rightarrow \tau = -\infty
 \end{aligned}$$

6. Time Scaling:

$$\begin{aligned}
 x(t) &\Leftrightarrow X(j\omega) \\
 x(at) &\Leftrightarrow \frac{1}{|a|} X\left(\frac{j\omega}{a}\right); a \neq 0 \\
 \text{proof :- } X(j\omega) &= \int_{-\infty}^{\infty} x(t) \cdot e^{-j\omega t} dt \\
 x(t) &\rightarrow x(at) \Rightarrow X(j\omega) \\
 \text{I) } a \text{ is positive} \\
 X(j\omega) &= \int_{-\infty}^{\infty} x(at) \cdot e^{-j\omega t} dt \\
 at = \tau &\Rightarrow t = \frac{\tau}{a} \Rightarrow dt = \frac{d\tau}{a}
 \end{aligned}$$

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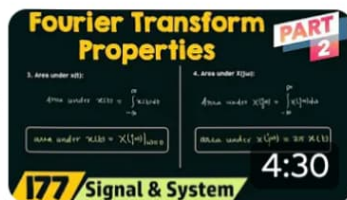
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Properties of Fourier Transform (Part 1)

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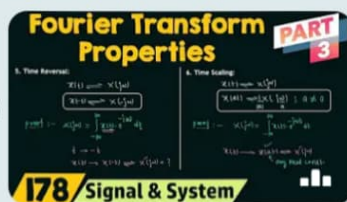
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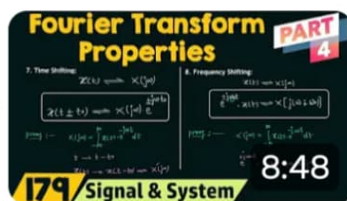
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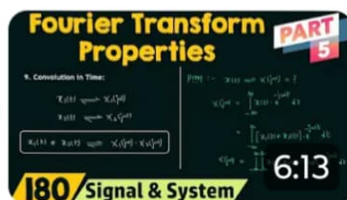
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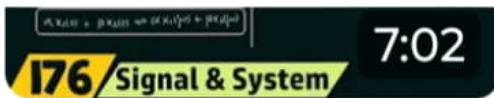
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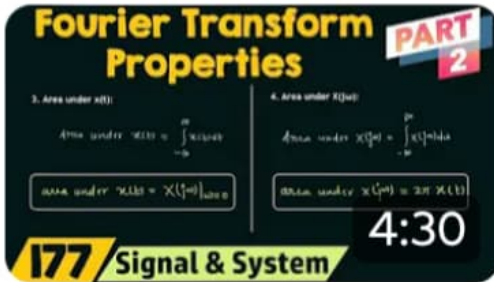


Properties of Fourier



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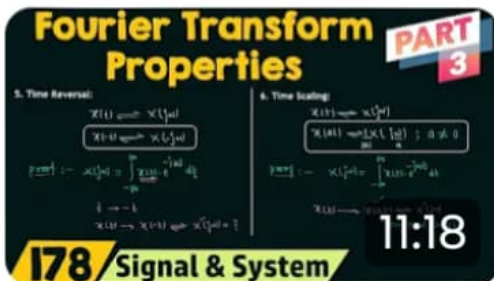
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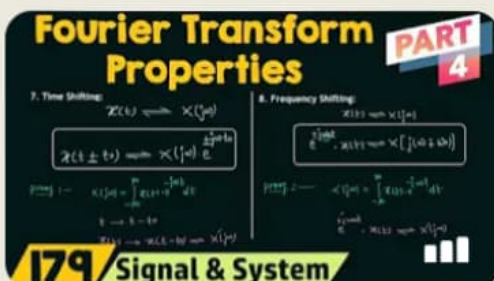
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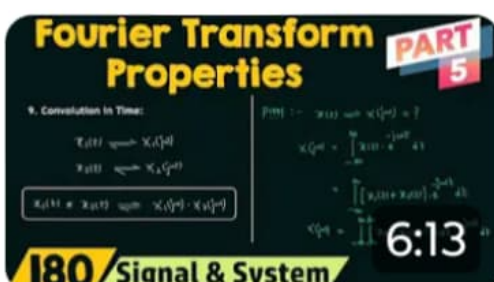
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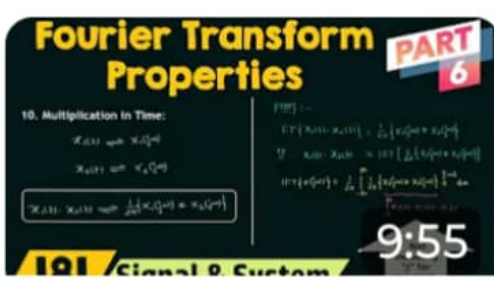
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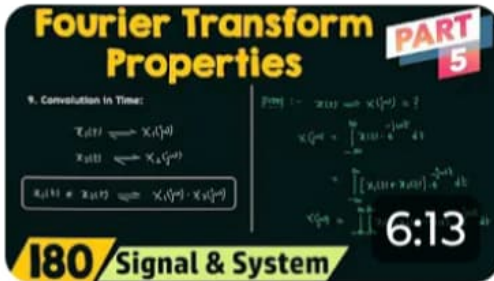
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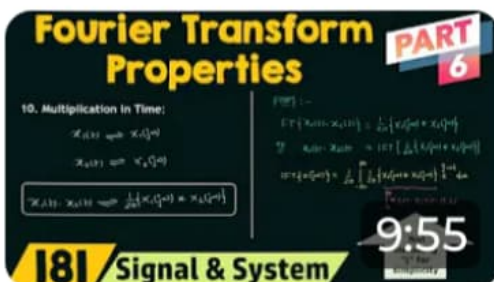
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Properties of Fourier Transform (Part 5)

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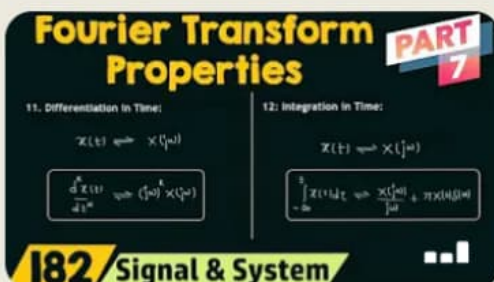
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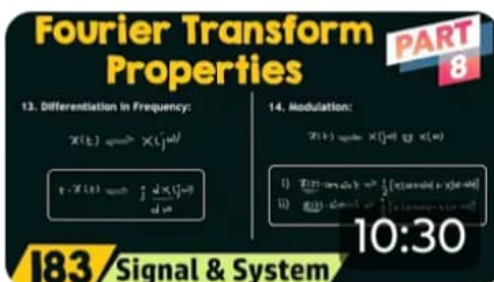
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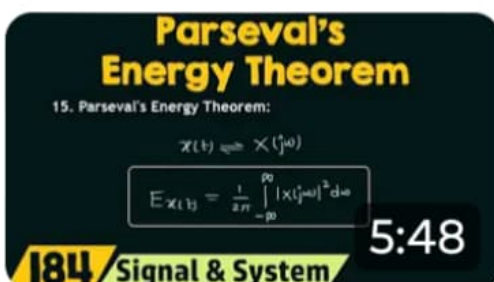
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Properties of Fourier Transform (Part 8)

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Parseval's Energy Theorem

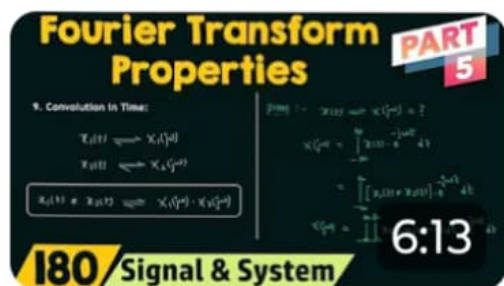
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Duality Property of Fourier

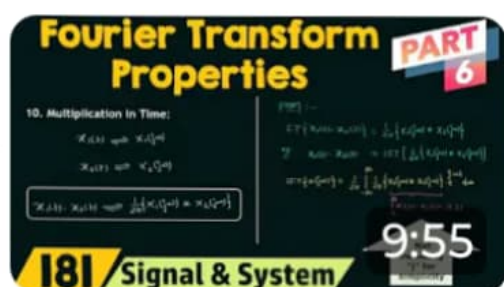




Properties of Fourier Transform (Part 5)

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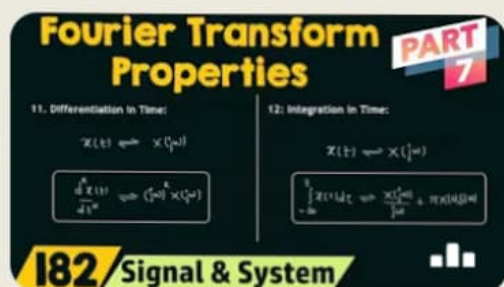
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Properties of Fourier Transform (Part 6)

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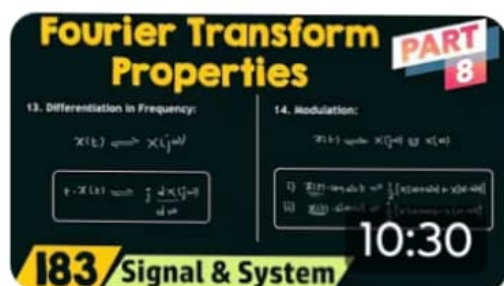
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Properties of Fourier Transf...

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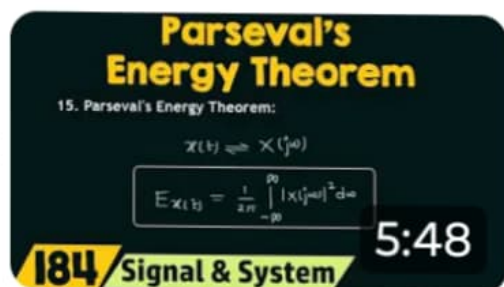
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Properties of Fourier Transform (Part 8)

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Parseval's Energy Theorem

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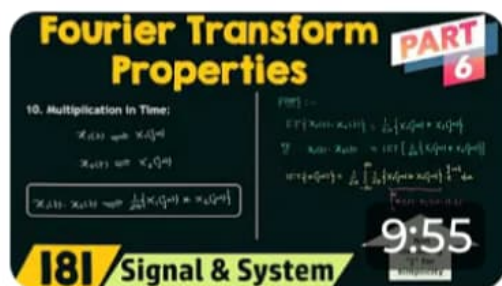
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Duality Property of Fourier



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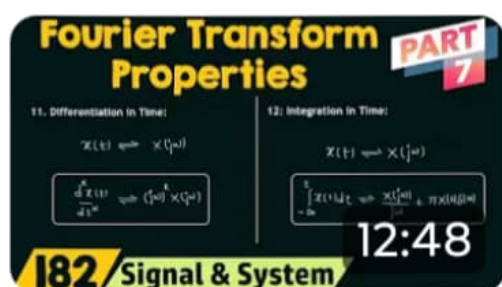
Properties of Fourier Transform (Part 6)

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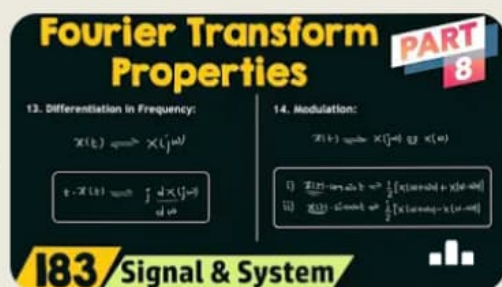
Properties of Fourier Transform (Part 7)

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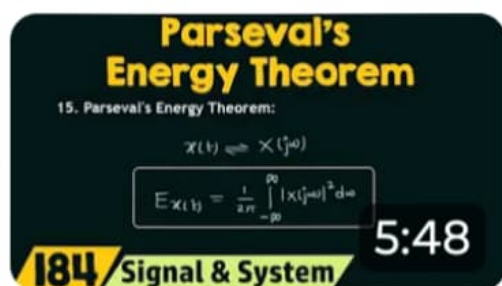
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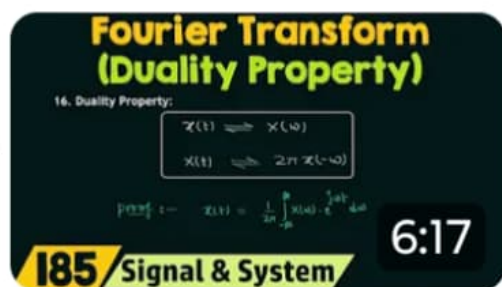
Parseval's Energy Theorem

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Duality Property of Fourier Transform

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Fourier Transform of Basic





Fourier Transform (Solved Problem 3)

Question: $x(t) = \frac{1}{a + jt} \Rightarrow X(\omega) = ?$

Homework: $y(t) = \frac{2a}{a^2 + t^2} \Rightarrow Y(\omega) = ?$

200 / Signal & System **2:45**

Fourier Transform (Solved Problem 3)

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Fourier Transform (Solved Problem 4)

Question: Find the Fourier transform of the signal shown below



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Fourier Transform (Solved Problem 4)

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Fourier Transform (Solved Problem 5)

Question: Find the Fourier Transform of the signal given below



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Fourier Transform (Solved P... :

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Fourier Transform (Solved Problem 6)

Question: Let $x(t) = \text{rect}(t - 1/2)$ [where $\text{rect}(t) = 1$ for $-1/2 \leq t \leq 1/2$ and zero otherwise]. Then if $\sin(t) = \sin(\pi t)/\pi t$, the Fourier Transform of $x(t) * x(-t)$ will be given by

- (a) $\sin(\omega/2\pi)$
- (b) $2\sin(\omega/2\pi)$
- (c) $2\sin(\omega/2\pi)\cos(\omega/2\pi)$
- (d) $\sin(\omega/2\pi)\sin(\omega/2\pi)$

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Fourier Transform (Solved Problem 6)

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Fourier Transform (Solved Problem 7)

Question: $x(t) * u(2t) = X(\omega) = ?$

$$u(t) = \begin{cases} 0 & t < 0 \\ 1 & t \geq 0 \end{cases} \quad \text{Homework: } X(\omega) = \frac{1}{j\omega} + \pi\delta(\omega)$$

$$u(1/2) = u(1/2) \Rightarrow u(1/2) = \frac{1}{2} + \pi\delta(\omega)$$

204 / Signal & System **4:03**

Fourier Transform (Solved Problem 7)

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Fourier Transform

Fourier Transform (Solved



Fourier Transform (Solved Problem 3)

Question: $x(t) = \frac{1}{a + jt} \Rightarrow X(\omega) = ?$

Homework: $y(t) = \frac{2a}{a^2 + t^2} \Rightarrow Y(\omega) = ?$

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2:45

Fourier Transform (Solved Problem 3)

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Fourier Transform (Solved Problem 4)

Question: Find the Fourier transform of the signal shown below



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Fourier Transform (Solved Problem 4)

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Fourier Transform (Solved Problem 5)

Question: Find the Fourier Transform of the signal given below



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Fourier Transform (Solved P...

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Fourier Transform (Solved Problem 6)

Question: Let $x(t) = \text{rect}(t - 1/2)$ [where $\text{rect}(t) = 1$ for $-1/2 \leq t \leq 1/2$ and zero otherwise]. Then if $\sin(t) = \sin(\omega t)/\omega$, the Fourier Transform of $x(t) * x(-t)$ will be given by

- (a) $\sin(\omega/2\pi)$
- (b) $2\sin(\omega/2\pi)$
- (c) $2\sin(\omega/2\pi)\cos(\omega/2)$
- (d) $\sin(\omega/2\pi)\sin(\omega/2)$

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Fourier Transform (Solved Problem 6)

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Fourier Transform (Solved Problem 7)

Question: $x(t) = u(2t) \Rightarrow X(\omega) = ?$

$$u(t) = \begin{cases} 1 & -1 < t < 1 \\ 0 & \text{otherwise} \end{cases} \Rightarrow \int_{-1}^1 e^{-j\omega t} dt = \frac{2 \sin(\omega/2)}{\omega}$$

Homework

$$u(t+1) = u(t) \Rightarrow \int_{-1}^1 e^{-j\omega t} dt = \frac{2 \sin(\omega/2)}{\omega}$$

4:03

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Fourier Transform (Solved Problem 7)

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Fourier Transform

Fourier Transform (Solved



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Fourier Transform (Solved Problem 2)

Question: If Fourier Transform of $x(t)$ is $X(\omega)$, then find the Fourier Transform of $y(t) = x(2t-3)$

Ans: 1- $X(\omega) \Rightarrow x(t)$
 $y(t) = x(2t-3) \Rightarrow Y(\omega) = ?$

Homework: $X(\omega) = X(\omega)$
 $y(t) = x(-3t+6) \Rightarrow Y(\omega) = ?$

PS: 1- $X(\omega) \xrightarrow{\text{Time Scaling}} X(\omega/2) \xrightarrow{\text{Time Shifting}} X(\omega/2) e^{-j\omega(3/2)}$
 PS: 2- $X(\omega) \xrightarrow{\text{Time Shifting}} X(\omega) e^{-j\omega(3/2)} \xrightarrow{\text{Time Scaling}} X(\omega/2) e^{-j\omega(3/2)}$

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Fourier Transform (Solved Problem 2)

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Fourier Transform (Solved Problem 3)

Question: $x(t) = \frac{1}{a + jt} \Rightarrow X(\omega) = ?$

Homework: $y(t) = \frac{2a}{a^2 + t^2} \Rightarrow Y(\omega) = ?$

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Fourier Transform (Solved Problem 3)

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Fourier Transform (Solved Problem 4)

Question: Find the Fourier transform of the signal shown below

Homework: $X(\omega) = X(\omega)$

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Fourier Transform (Solved P...

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Fourier Transform (Solved Problem 5)

Question: Find the Fourier Transform of the signal given below

Homework: $X(\omega) = X(\omega)$

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Fourier Transform (Solved Problem 5)

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Fourier Transform (Solved Problem 6)

Question: Let $x(t) = \text{rect}(t-1/2)$ [where $\text{rect}(t) = 1$ for $-1/2 \leq t \leq 1/2$ and zero otherwise]. Then if $\sin(\omega t) = \sin(\omega t)/\omega t$, the Fourier Transform of $x(t) * x(-t)$ will be given by

(a) $\text{sinc}(\omega/2)$
 (b) $2\text{sinc}(\omega/2)$
 (c) $2\text{sinc}(\omega/2) \cos(\omega/2)$
 (d) $\text{sinc}(\omega/2) \cos(\omega/2)$

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Fourier Transform (Solved Problem 6)

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Fourier Transform

Fourier Transform (Solved





Fourier Transform (Solved Problem 1)

Question: $x(t) = e^{-at^2}$, $a > 0$
 $X(\omega) = ?$

$\int_{-\infty}^{\infty} e^{-at^2} e^{j\omega t} dt = \sqrt{\frac{\pi}{a}} e^{-\omega^2/4a}$

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10:09

Gaussian

Fourier Transform (Solved Problem 1)

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Fourier Transform (Solved Problem 2)

Question: If Fourier Transform of $x(t)$ is $X(\omega)$, then find the Fourier Transform of $y(t) = x(2t-3)$

Ans: $y(t) = x(2t-3) \Rightarrow Y(\omega) = ?$

Homework: $x(t) = X(\omega)$
 $y(t) = x(2t-3) \Rightarrow Y(\omega) = ?$

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Fourier Transform (Solved Problem 2)

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Fourier Transform (Solved Problem 3)

Question: $x(t) = \frac{1}{a + jt} \Rightarrow X(\omega) = ?$

Homework: $y(t) = \frac{2a}{a^2 + t^2} \Rightarrow Y(\omega) = ?$

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Fourier Transform (Solved P...

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Fourier Transform (Solved Problem 4)

Question: Find the Fourier transform of the signal shown below

$x(t)$ and $X(\omega)$ plots are shown.

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Fourier Transform (Solved Problem 4)

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Fourier Transform (Solved Problem 5)

Question: Find the Fourier Transform of the signal given below

$x(t)$ and $X(\omega)$ plots are shown.

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Fourier Transform (Solved Problem 5)

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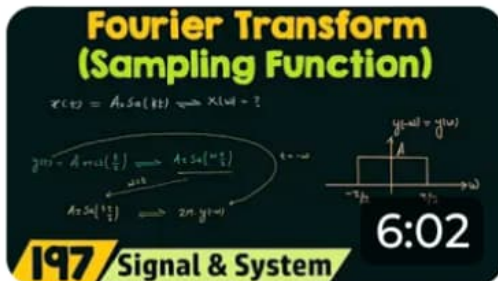
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Fourier Transform

Fourier Transform (Solved

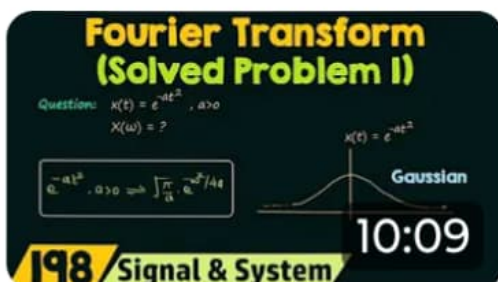




Fourier Transform of Basic Signals (Sampling Function)

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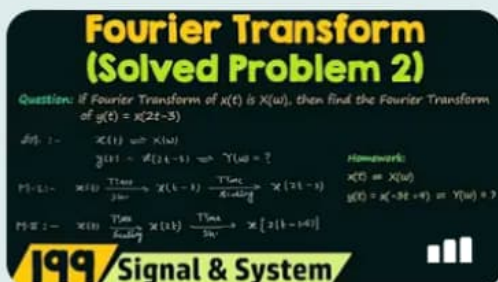
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Fourier Transform (Solved Problem 1)

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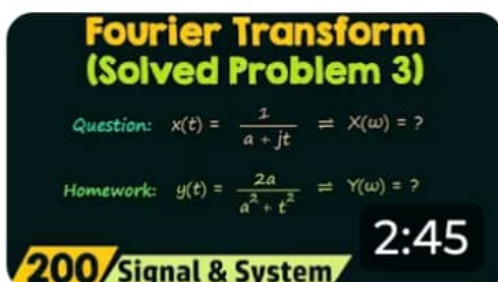
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Fourier Transform (Solved P...

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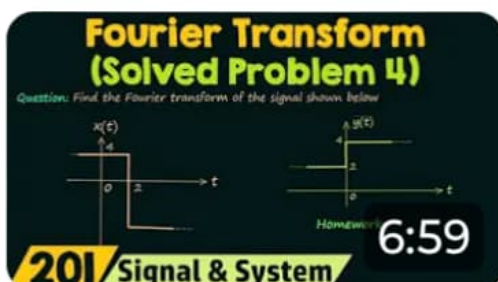
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Fourier Transform (Solved Problem 3)

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Fourier Transform (Solved Problem 4)

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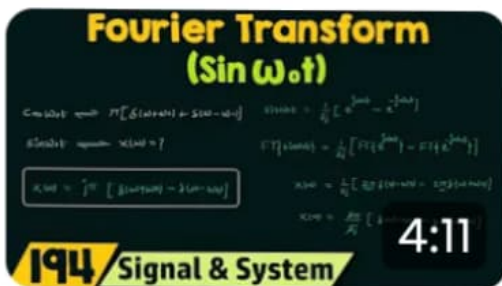


Fourier Transform (Solved Problem 5)





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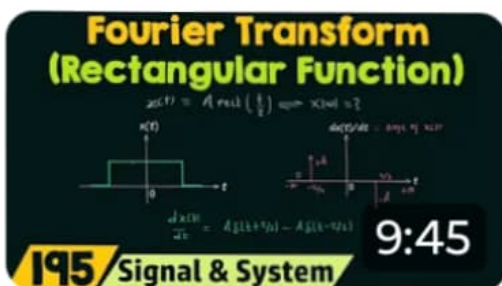
Fourier Transform of Basic Signals (Sin $\omega_0 t$)

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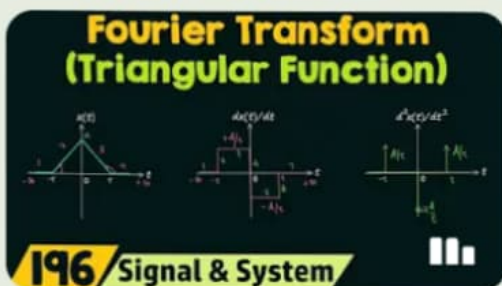
Fourier Transform of Basic Signals (Rectangular F...

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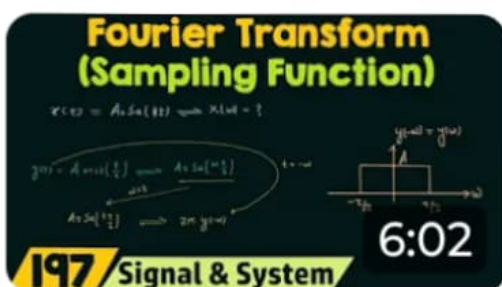
Fourier Transform of Basic ...

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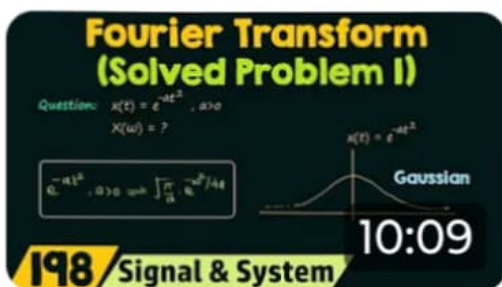
Fourier Transform of Basic Signals (Sampling Function)

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Fourier Transform (Solved Problem 1)

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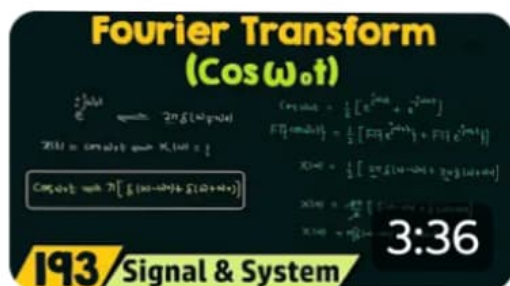


Fourier Transform (Solved





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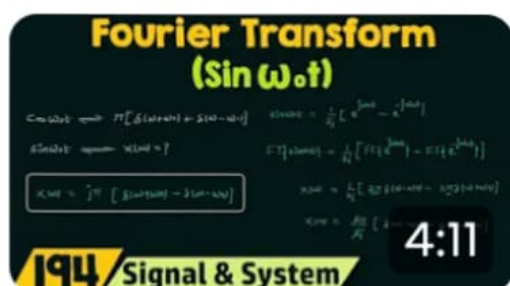
Fourier Transform of Basic Signals (Cos $\omega_0 t$)

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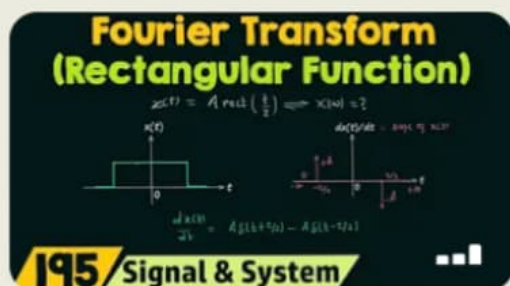
Fourier Transform of Basic Signals (Sin $\omega_0 t$)

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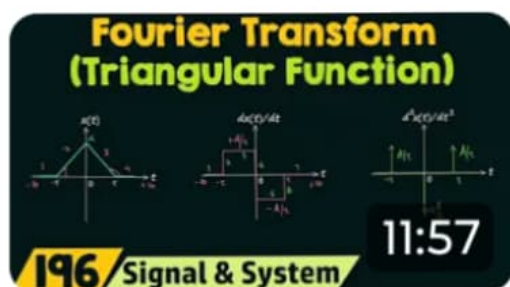
Fourier Transform of Basic ...

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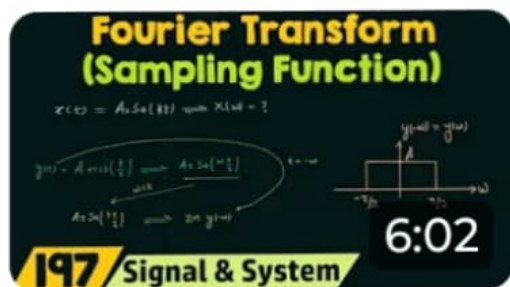
Fourier Transform of Basic Signals (Triangular Function)

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Fourier Transform of Basic Signals (Sampling Function)

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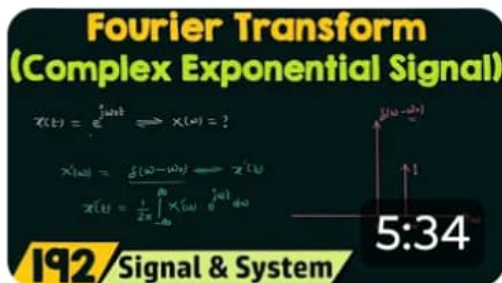


Fourier Transform (Solved)





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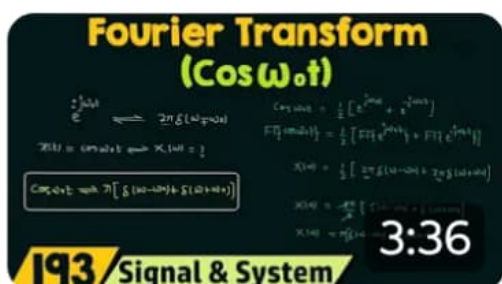
Fourier Transform of Basic Signals (Complex Exp...

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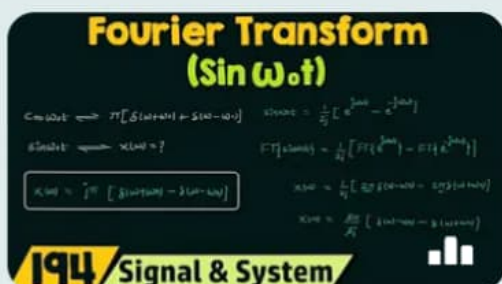
Fourier Transform of Basic Signals (Cos\omega_0 t)

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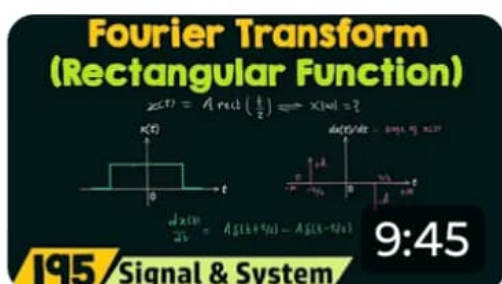
Fourier Transform of Basic ...

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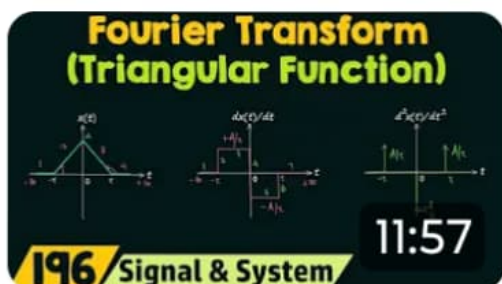
Fourier Transform of Basic Signals (Rectangular F...

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Fourier Transform of Basic Signals (Triangular Function)

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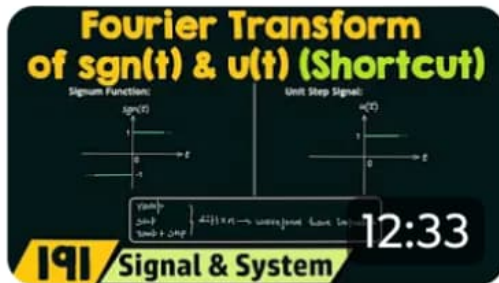


Fourier Transform of Basic





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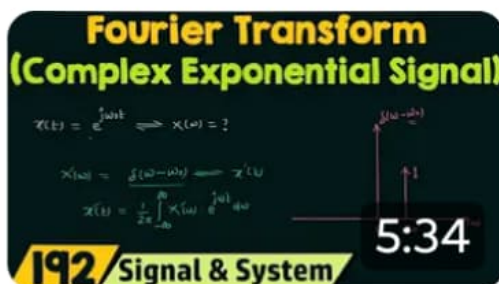


Fourier Transform of $\text{sgn}(t)$ & $u(t)$ [Important Shortcut]

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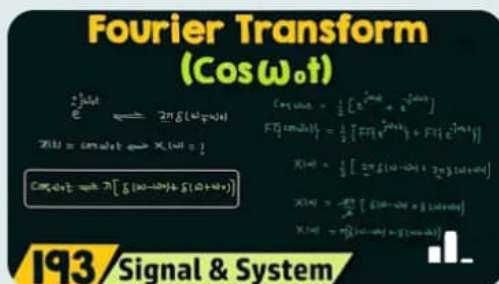


Fourier Transform of Basic Signals (Complex Exp...)

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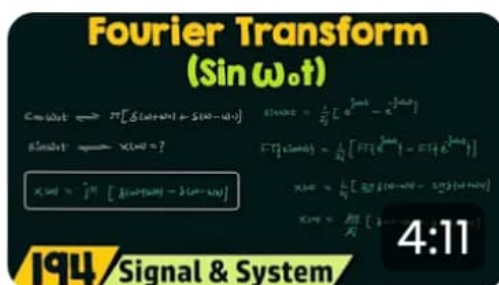
Fourier Transform of Basic ...

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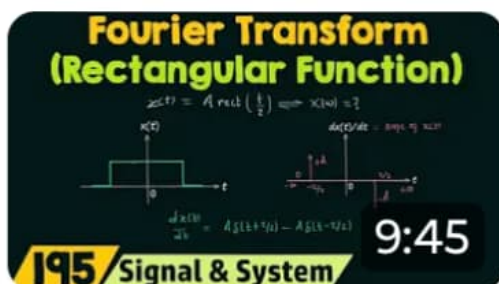


Fourier Transform of Basic Signals ($\sin \omega_0 t$)

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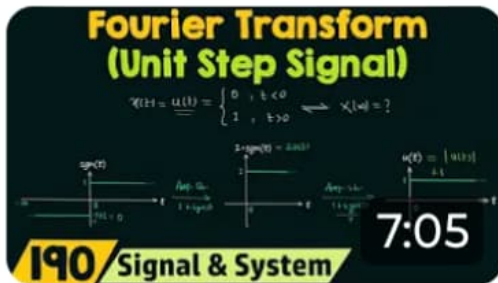
Fourier Transform of Basic Signals (Rectangular F...)

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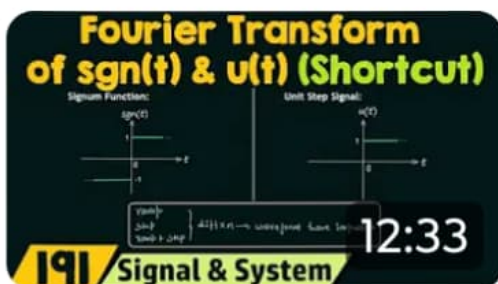
Fourier Transform of Basic Signals (Triangular F...)



Fourier Transform of Basic Signals (Unit Step Signal)

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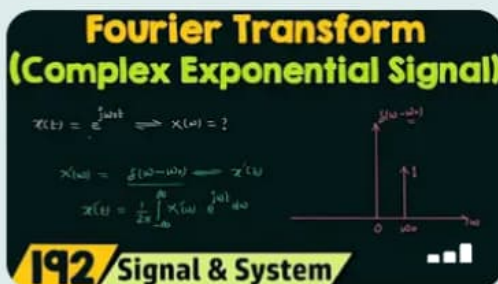
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Fourier Transform of sgn(t) & u(t) [Important Shortcut]

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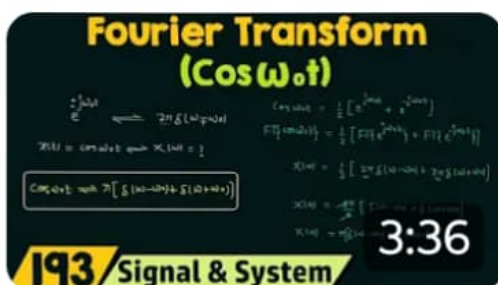
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Fourier Transform of Basic ...

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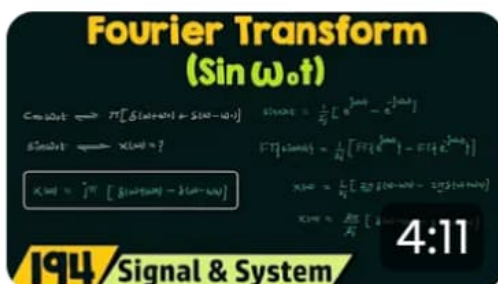
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Fourier Transform of Basic Signals (Cosω_0 t)

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Fourier Transform of Basic Signals (Sinω_0 t)

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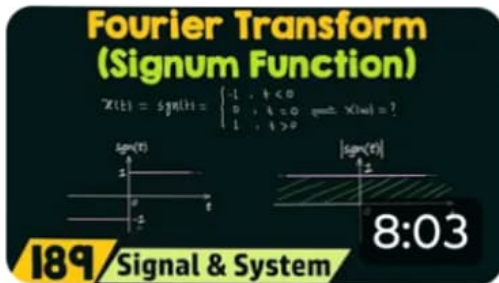


Fourier Transform of





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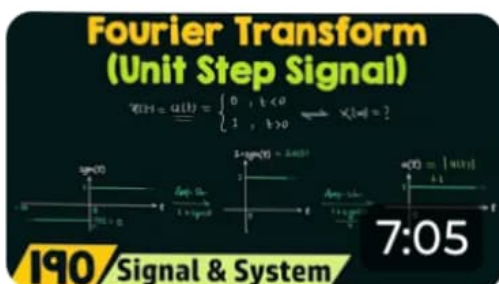
Fourier Transform of Basic Signals (Signum Function)

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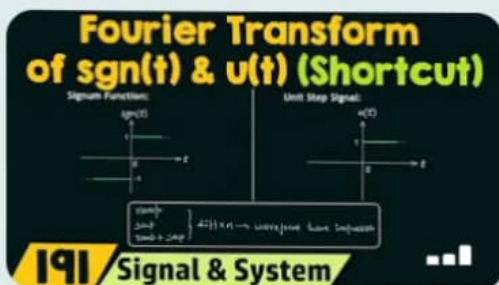
Fourier Transform of Basic Signals (Unit Step Signal)

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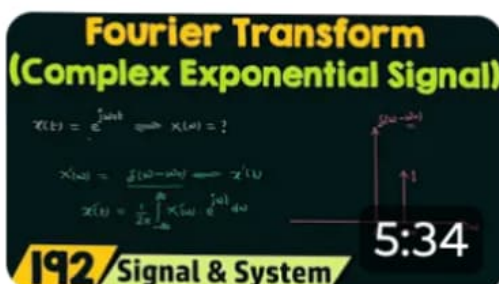
Fourier Transform of sgn(t) ...

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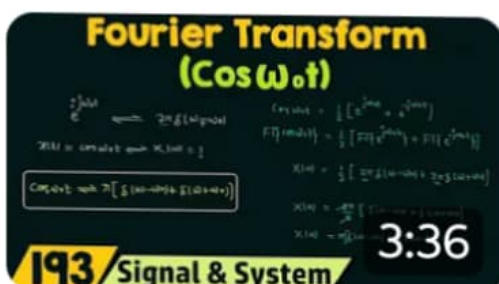
Fourier Transform of Basic Signals (Complex Exp...

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Fourier Transform of Basic Signals (Cos\omega_0 t)

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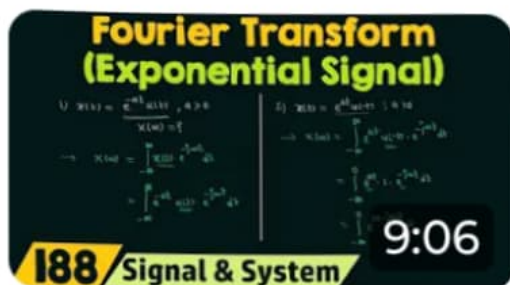


Fourier Transform of Basic Signals (Sine Wave)





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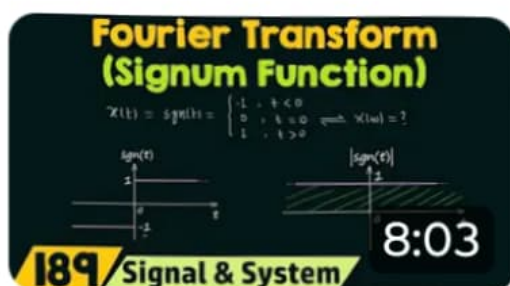
Fourier Transform of Basic Signals (Exponential Signals)

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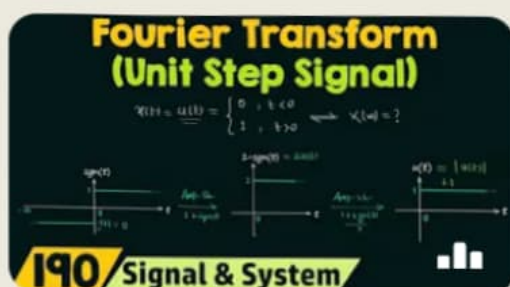
Fourier Transform of Basic Signals (Signum Function)

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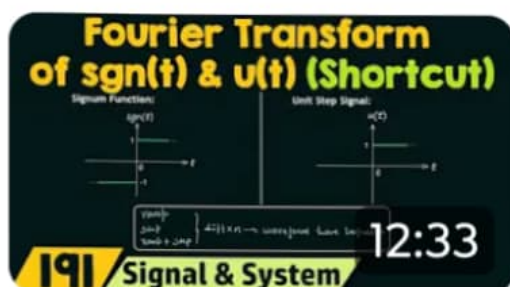
Fourier Transform of Basic ...

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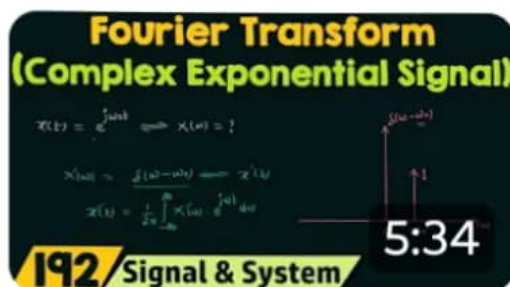
Fourier Transform of $\text{sgn}(t)$ & $u(t)$ [Important Shortcut]

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Fourier Transform of Basic Signals (Complex Exp...)

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Fourier Transform of Basic





Fourier Transform (Impulse Signal)

$x(t) = \delta(t) \Rightarrow X(\omega) = 1$

$X(\omega) = \int_{-\infty}^{\infty} \delta(t) e^{-j\omega t} dt = 1$

Impulse related signals are absolutely integrable signals and hence we can use the formula of Fourier Transform

187 / Signal & System **3:16**

Fourier Transform of Basic Signals (Impulse Signal)

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Fourier Transform (Exponential Signal)

1) $x(t) = e^{at} u(t), a > 0$

$X(\omega) = \int_{-\infty}^{\infty} e^{at} u(t) e^{-j\omega t} dt = \int_0^{\infty} e^{(a-j\omega)t} dt = \frac{1}{a-j\omega}$

2) $x(t) = e^{at} u(-t), a > 0$

$X(\omega) = \int_{-\infty}^{\infty} e^{at} u(-t) e^{-j\omega t} dt = \int_{-\infty}^0 e^{(a-j\omega)t} dt = \frac{1}{a-j\omega}$

188 / Signal & System **9:06**

Fourier Transform of Basic Signals (Exponential Signals)

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Fourier Transform (Signum Function)

$x(t) = \text{sgn}(t) = \begin{cases} -1, & t < 0 \\ 0, & t = 0 \\ 1, & t > 0 \end{cases} \Rightarrow X(\omega) = ?$

$X(\omega) = \int_{-\infty}^{\infty} \text{sgn}(t) e^{-j\omega t} dt = \int_{-\infty}^0 -e^{-j\omega t} dt + \int_0^{\infty} e^{-j\omega t} dt = \frac{1}{j\omega}$

189 / Signal & System

Fourier Transform of Basic ...

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Fourier Transform (Unit Step Signal)

$x(t) = u(t) = \begin{cases} 0, & t < 0 \\ 1, & t > 0 \end{cases} \Rightarrow X(\omega) = ?$

$X(\omega) = \int_{-\infty}^{\infty} u(t) e^{-j\omega t} dt = \int_0^{\infty} e^{-j\omega t} dt = \frac{1}{j\omega} + \pi \delta(\omega)$

190 / Signal & System **7:05**

Fourier Transform of Basic Signals (Unit Step Signal)

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Fourier Transform of $\text{sgn}(t)$ & $u(t)$ (Shortcut)

Signum Function: $\text{sgn}(t) = \frac{1}{j\omega}$

Unit Step Signal: $u(t) = \frac{1}{j\omega} + \pi \delta(\omega)$

191 / Signal & System **12:33**

Fourier Transform of $\text{sgn}(t)$ & $u(t)$ [Important Shortcut]

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Fourier Transform

Fourier Transform of

$$X(\omega) = ?$$

$$X(\omega) = \int_{-\infty}^{\infty} \underline{x(t)} \cdot e^{-j\omega t} dt$$

$$= \int_{-\infty}^{\infty} e^{-at} \underline{u(t)} \cdot e^{-j\omega t} dt$$

$$= \int_0^{\infty} \underline{e^{-at}} \cdot \underline{1} \cdot \underline{e^{-j\omega t}} dt$$

$$= \int_0^{\infty} e^{-(a+j\omega)t} dt$$

$$= \left[\frac{e^{-(a+j\omega)t}}{-(a+j\omega)} \right]_0^{\infty}$$

$$= \frac{-1}{a+j\omega} \left[\cancel{e^{-\infty}} - \cancel{e^0} \right]$$

$\frac{1}{\infty} = \frac{1}{\infty} = 0$

$$\rightarrow X(\omega) = \int_{-\infty}^{\infty} e^{at} \underline{u(-t)} \cdot e^{-j\omega t} dt$$

$$= \int_{-\infty}^0 \underline{e^{at}} \cdot \underline{1} \cdot \underline{e^{-j\omega t}} dt$$

$$= \int_{-\infty}^0 e^{(a-j\omega)t} dt$$

$$= \left[\frac{e^{(a-j\omega)t}}{a-j\omega} \right]_{-\infty}^0$$

$$= \frac{1}{a-j\omega} \left[\cancel{e^0} - \cancel{e^{-\infty}} \right]$$

$$X(\omega) = \frac{1}{a-j\omega}$$

$$e^{at} u(-t) \Rightarrow \frac{1}{a-j\omega} ; a > 0$$

$$\rightarrow X(\omega) = \int_{-\infty}^{\infty} e^{-a|t|} \cdot e^{-j\omega t} dt$$

$$= \int_{-\infty}^0 e^{at} \cdot e^{-j\omega t} dt +$$

$$\int_0^{\infty} e^{-at} \cdot e^{-j\omega t} dt$$

$$= \frac{1}{a-j\omega} + \frac{1}{a+j\omega}$$

$$= \frac{\cancel{a+j\omega} + \cancel{a-j\omega}}{a^2 - j^2\omega^2}$$

$$X(\omega) = \frac{2a}{a^2 + \omega^2}$$

$$e^{-a|t|} \Rightarrow \frac{2a}{a^2 + \omega^2} ; a > 0$$

$$X(\omega) = ?$$

$$X(\omega) = \int_{-\infty}^{\infty} \underline{x(t)} \cdot e^{-j\omega t} dt$$

$$= \int_{-\infty}^{\infty} e^{-at} \underline{u(t)} \cdot e^{-j\omega t} dt$$

$$= \int_0^{\infty} \underline{e^{-at}} \cdot \underline{1} \cdot \underline{e^{-j\omega t}} dt$$

$$= \int_0^{\infty} e^{-(a+j\omega)t} dt$$

$$= \left[\frac{e^{-(a+j\omega)t}}{-(a+j\omega)} \right]_0^{\infty}$$

$$= \frac{-1}{a+j\omega} \left[\cancel{e^{-\infty}} - \cancel{e^0} \right]$$

$$\frac{1}{\infty} = \frac{1}{\infty} = 0$$

$$\rightarrow X(\omega) = \int_{-\infty}^{\infty} e^{at} \underline{u(-t)} \cdot e^{-j\omega t} dt \rightarrow X(\omega) = \int_{-\infty}^0 e^{-a|t|} \cdot e^{-j\omega t} dt$$

$$= \int_{-\infty}^0 \underline{e^{at}} \cdot \underline{1} \cdot \underline{e^{-j\omega t}} dt$$

$$= \int_{-\infty}^0 e^{(a-j\omega)t} dt$$

$$= \left[\frac{e^{(a-j\omega)t}}{a-j\omega} \right]_{-\infty}^0$$

$$= \frac{1}{a-j\omega} \left[\cancel{e^0} - \cancel{e^{-\infty}} \right]$$

$$X(\omega) = \frac{1}{a-j\omega}$$

$$e^{at} u(-t) \Rightarrow \frac{1}{a-j\omega} ; a > 0$$

$$= \int_{-\infty}^0 e^{at} \cdot e^{-j\omega t} dt +$$

$$\int_0^{\infty} e^{-at} \cdot e^{-j\omega t} dt$$

$$= \frac{1}{a-j\omega} + \frac{1}{a+j\omega}$$

$$= \frac{a+j\omega + a-j\omega}{a^2 - j^2\omega^2}$$

$$X(\omega) = \frac{2a}{a^2 + \omega^2} \quad \leftarrow -1$$

$$e^{-a|t|} \Rightarrow \frac{2a}{a^2 + \omega^2} ; a > 0$$

Fourier Transform of Basic Signals (Exponential Signals)

$$x(t) = \underline{e^{-at} u(t)}, \quad a > 0$$

$$X(\omega) = ?$$

$$\begin{aligned} X(\omega) &= \int_{-\infty}^{\infty} \underline{x(t)} \cdot e^{-j\omega t} dt \\ &= \int_{-\infty}^{\infty} e^{-at} \underline{u(t)} \cdot e^{-j\omega t} dt \\ &= \int_0^{\infty} \underline{e^{-at} \cdot 1 \cdot e^{-j\omega t}} dt \\ &= \int_0^{\infty} e^{-(a+j\omega)t} dt \\ &= \left[\frac{e^{-(a+j\omega)t}}{-(a+j\omega)} \right]_0^{\infty} \end{aligned}$$

$$\text{ii) } x(t) = \underline{e^{at} u(-t)}; \quad a > 0$$

$$\rightarrow X(\omega) = \int_{-\infty}^{\infty} e^{at} \underline{u(-t)} \cdot e^{-j\omega t} dt$$

$$= \int_{-\infty}^0 \underline{e^{at} \cdot 1 \cdot e^{-j\omega t}} dt$$

$$= \int_{-\infty}^0 e^{(a-j\omega)t} dt$$

$$= \left[\frac{e^{(a-j\omega)t}}{a-j\omega} \right]_{-\infty}^0$$

$$= \frac{1}{a-j\omega} \left[\underset{\substack{\nearrow 1 \\ \nwarrow 0}}{e^0} - \underset{\substack{\nearrow 0 \\ \nwarrow 1}}{e^{-\infty}} \right]$$

$$X(\omega) = \frac{1}{a-j\omega}$$

$$\text{iii) } x(t) = \underline{e^{-a|t|}}; \quad a > 0$$

$$\rightarrow X(\omega) = \int_{-\infty}^{\infty} e^{-a|t|} \cdot e^{-j\omega t} dt$$

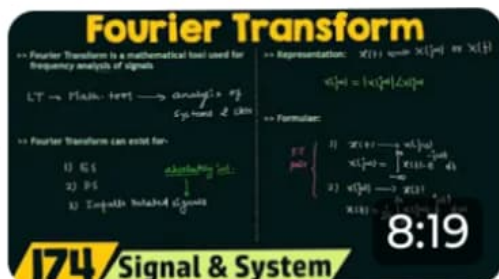
$$= \int_{-\infty}^0 e^{at} \cdot e^{-j\omega t} dt + \int_0^{\infty} e^{-at} \cdot e^{-j\omega t} dt$$

$$= \int_0^{\infty} e^{-at} \cdot e^{-j\omega t} dt + \int_0^{\infty} e^{-at} \cdot e^{j\omega t} dt$$

$$= \frac{1}{a-j\omega} + \frac{1}{a+j\omega}$$

$$= \frac{a+j\omega + a-j\omega}{a^2 - j^2\omega^2}$$

$$X(\omega) = \frac{2a}{a^2 + \omega^2}$$



Introduction to Fourier Transform

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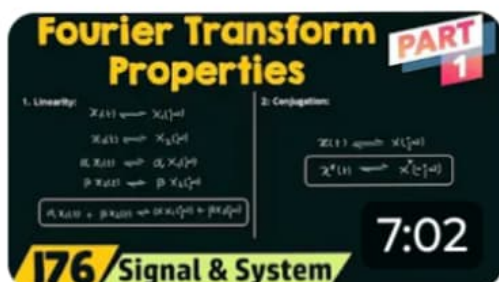
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Conditions for Existence of Fourier Transform (Dirichlet...

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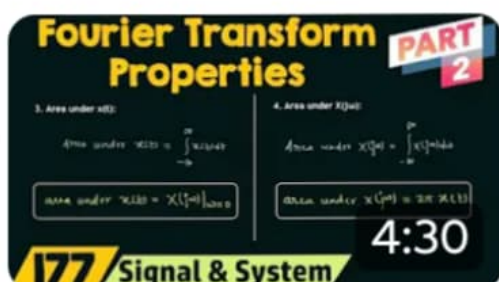
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Properties of Fourier Transform (Part 1)

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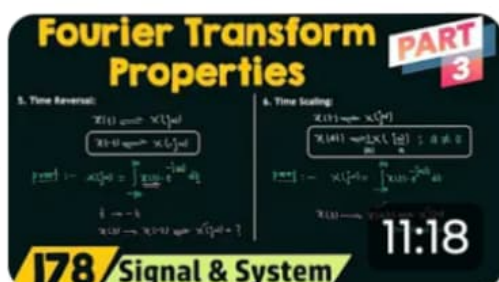
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Properties of Fourier Transform (Part 2)

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Properties of Fourier Transform (Part 3)

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Properties of Fourier Transform (Part 4)





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Parseval's Energy Theorem

15. Parseval's Energy Theorem:

$$x(t) \rightleftharpoons X(j\omega)$$

$$E_{x(t)} = \frac{1}{2\pi} \int_{-\infty}^{\infty} |X(j\omega)|^2 d\omega$$

184 / Signal & System **5:48**

Parseval's Energy Theorem

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Fourier Transform (Duality Property)

16. Duality Property:

$$x(t) \rightleftharpoons X(j\omega)$$

$$X(t) \rightleftharpoons 2\pi x(-j\omega)$$

proof: $\therefore x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega$

185 / Signal & System **6:17**

Duality Property of Fourier Transform

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Fourier Transform (DC Value)

$$x(t) \rightleftharpoons X(j\omega) = A_0 \delta(\omega)$$

$$X(j\omega) = \frac{1}{2\pi} \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} A_0 \delta(t) e^{-j\omega t} dt$$

$$= \frac{A_0}{2\pi} \int_{-\infty}^{\infty} \delta(t) dt = \frac{A_0}{2\pi}$$

$$A_0 \rightleftharpoons 2\pi A_0 \delta(\omega)$$

186 / Signal & System

Fourier Transform of Basic ...

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Fourier Transform (Impulse Signal)

$$x(t) = \delta(t) \rightleftharpoons X(j\omega) = 1$$

$$X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

$$= \int_{-\infty}^{\infty} \delta(t) e^{-j\omega t} dt$$

$$= \int_{-\infty}^{\infty} \delta(t) dt = 1$$

Impulse related signals are absolutely integrable signals and hence we can use the formula of Fourier Transform

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Fourier Transform of Basic Signals (Impulse Signal)

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Fourier Transform (Exponential Signal)

$$1) x(t) = e^{at} u(t), a > 0$$

$$X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

$$= \int_{0}^{\infty} e^{at} e^{-j\omega t} dt$$

$$= \int_{0}^{\infty} e^{(a-j\omega)t} dt$$

$$= \left[\frac{e^{(a-j\omega)t}}{a-j\omega} \right]_{0}^{\infty}$$

$$= \frac{1}{a-j\omega}$$

$$2) x(t) = e^{at} u(-t), a > 0$$

$$X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

$$= \int_{-\infty}^{0} e^{at} e^{-j\omega t} dt$$

$$= \int_{-\infty}^{0} e^{(a-j\omega)t} dt$$

$$= \left[\frac{e^{(a-j\omega)t}}{a-j\omega} \right]_{-\infty}^{0}$$

$$= \frac{1}{a-j\omega}$$

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Fourier Transform of Basic Signals (Exponential Signals)

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Fourier Transform

Fourier Transform of Basic



$$\int_{-\infty}^{\infty} |x(t)| dt < \infty$$

$$\int_{-\infty}^{\infty} A_0 dt = A_0 \int_{-\infty}^{\infty} dt$$

$$= A_0 (t)_{-\infty}^{\infty}$$

$$= A_0 [\infty - (-\infty)]$$

$$= A_0 (\infty + \infty)$$

$$= A_0 \times \infty$$

$$= \infty$$

having the Fourier transform of signal a
naught which

$$x(t) \Rightarrow X(\omega) \text{ or } X(\omega)$$

$$X(\omega) = \int_{-\infty}^{\infty} x(t) \cdot e^{-j\omega t} dt$$

$$= \int_{-\infty}^{\infty} A_0 \cdot e^{-j\omega t} dt$$

$$= A_0 \int_{-\infty}^{\infty} e^{-j\omega t} dt$$

$$= A_0 \left(\frac{e^{-j\omega t}}{-j\omega} \right)_{-\infty}^{\infty}$$

$$= \frac{A_0}{-j\omega} (e^{-j\omega \infty} - e^{-j\omega (-\infty)})$$

$$= \frac{A_0}{\omega} \left(\frac{e^{j\omega \infty} - e^{j\omega (-\infty)}}{j} \right)$$

$$X(\omega) = A_0 \cdot 2 \sin(\omega)$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) \cdot e^{j\omega t} d\omega$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} A_0 \delta(\omega) \cdot e^{j\omega t} d\omega$$

$$= \frac{A_0}{2\pi} \int_{-\infty}^{\infty} \underbrace{\delta(\omega - 0)}_{\text{Area}} \cdot \underbrace{e^{j\omega t}}_{\text{Area}} d\omega$$

$$= \frac{A_0}{2\pi} \int_{-\infty}^{\infty} \delta(\omega) d\omega$$

1 = Area

$$x(t) = \frac{A_0}{2\pi}$$

$$\frac{A_0}{2\pi} \Rightarrow A_0 \delta(\omega)$$

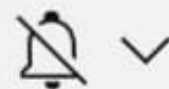
$$A_0 \Rightarrow 2\pi A_0 \delta(\omega)$$

Duality Property of Fourier Transform

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